

CLASS- X
MARKING SCHEME: MATHEMATICS
PB SET 1

	<u>Section A</u>	
1.	(a) 1260	1
2.	(b) 1 zero & the zero is 3	1
3.	(d) No solution	1
4.	(b) $\pm 8/5$	1
5.	(c) 7	1
6.	(a) 6	1
7.	(a) 30 cm	1
8.	(b) $\angle A = \angle E$	1
9.	(a) 2.4 cm	1
10.	(b) 70°	1
11.	(b) 100°	1
12.	(c) 28 cm	1
13.	(d) $\cos A$	1
14.	(b) 9	1
15.	(c) 1694 m^2	1
16.	(b) Median	1
17.	(c) 13	1
18.	(c) $\frac{3}{4}$	1
19.	(b) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)	1
20.	(d) Assertion (A) is false but reason (R) is true	1
	<u>Section B</u>	

21.	(a) $4 \times 11 \times 13 + 4$ $= 4 \times (11 \times 13 + 1)$ $= 4 \times 144 = 2 \times 2 \times 2 \times 2 \times 3 \times 2 \times 2 \times 3$ Since, $4 \times 11 \times 13 + 4$ has factors other than 1 & itself so, it is a composite no. OR (b) If the number 4^n ends with the digit zero (0), then it should be divisible by 5, as we know any number with unit place as 0 or 5 is divisible by 5. Prime factorization of $4^n = (2 \times 2)^n$ Therefore, the prime factorization of 4^n doesn't contain prime number 5. Hence, it is clear that for any natural number n, 4^n is not divisible by 5, and thus it proves that 4^n cannot end with the digit 0 for any natural number n.	1 1 1 1
22.	$\alpha + \beta = k - 6$, $\alpha\beta = 2(2k - 1)$ $\alpha + \beta = \alpha\beta / 2$ $k - 6 = 2(2k - 1) / 2$ $k - 6 = 2k - 1$ $-6 + 1 = 2k - k$, $k = -5$	1 1
23.	$3x^2 - 4\sqrt{3}x + 4 = 0$ $a=3$, $b= -4\sqrt{3}$, $c = 4$ $D = b^2 - 4ac$ $= (-4\sqrt{3})^2 - 4 \times 3 \times 4 = 48 - 48 = 0$ The equation has real and equal roots.	1 1
24.	(a) $\sin^2 60^\circ + 2 \tan 45^\circ - \cos^2 30^\circ$ $= (\sqrt{3}/2)^2 + 2 \times 1 - (\sqrt{3}/2)^2$ $= 2$ OR (b) $\sin x = 7/25$ $\cos x = 24/25$ $\tan x = 7/24$ $\cot x = 24/7$ $\sin x \cos x (\tan x + \cot x)$ $= 7/25 \times 24/25 (7/24 + 24/7)$ $= 7/25 \times 24/25 \times 625/168$ $= 1$	1 1 1 1

25.	$P(x, y) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$ $x = \frac{1(3) + 2(7)}{1+2} = \frac{3+14}{3} = \frac{17}{3}$ $y = \frac{1(4) + 2(-6)}{1+2} = \frac{4-12}{3} = \frac{-8}{3}$ The point is (17/3, -8/3)	1
	<u>Section C</u>	1

26.	Let us assume, that $\sqrt{5}$ is rational number. i.e. $\sqrt{5} = x/y$ (where, x and y are co-primes, integers, $y \neq 0$) $y\sqrt{5} = x$ Squaring both the sides, we get, $(y\sqrt{5})^2 = x^2$ $\Rightarrow 5y^2 = x^2 \dots\dots\dots (1)$ Thus, x^2 is divisible by 5, so x is also divisible by 5. Let us say, $x = 5k$, for some value of k and substituting the value of x in equation (1), we get, $5y^2 = (5k)^2$ $\Rightarrow y^2 = 5k^2$ is divisible by 5 it means y is divisible by 5. Clearly, x and y are not co-primes. Thus, our assumption about $\sqrt{5}$ is rational is incorrect. Hence, $\sqrt{5}$ is an irrational number. Let us assume $3 + 2\sqrt{5}$ is rational. Then we can find co-prime x and y ($y \neq 0$) such that $3 + 2\sqrt{5} = x/y$ Rearranging, we get, $2\sqrt{5} = \frac{x}{y} - 3$ $\sqrt{5} = \frac{1}{2} \left(\frac{x}{y} - 3 \right)$ Since, x and y are integers, thus, $\frac{1}{2} \left(\frac{x}{y} - 3 \right)$ is a rational number.	$\frac{1}{2}$ 1 1
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Therefore, $\sqrt{5}$ is also a rational number. But this contradicts the fact that $\sqrt{5}$ is irrational.

So, we conclude that $3 + 2\sqrt{5}$ is irrational.

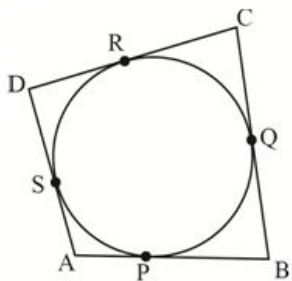
$\frac{1}{2}$

27.	First term, $a = 5$ and common difference, $d = 1.75$ Also given, $a_n = 20.75$ Find, $n = ?$ As we know, by the nth term formula, $a_n = a + (n-1)d$ Therefore, $20.75 = 5 + (n-1) \times 1.75$ $15.75 = (n-1) \times 1.75$ $(n-1) = 15.75 / 1.75 = 1575 / 175$ $= 63 / 7 = 9$ $n-1 = 9$ $n = 10$ Hence, n is 10.	1 1 1
28.	$\tan \theta / (1 - \cot \theta) + \cot \theta / (1 - \tan \theta) = 1 + \sec \theta \operatorname{cosec} \theta$ $= [(\sin \theta / \cos \theta) / (1 - (\cos \theta / \sin \theta))] + [(\cos \theta / \sin \theta) / (1 - (\sin \theta / \cos \theta))]$ $= [(\sin \theta / \cos \theta) / ((\sin \theta - \cos \theta) / \sin \theta)] + [(\cos \theta / \sin \theta) / ((\cos \theta - \sin \theta) / \cos \theta)]$ $= \sin^2 \theta / [\cos \theta (\sin \theta - \cos \theta)] + \cos^2 \theta / [\sin \theta (\cos \theta - \sin \theta)]$	1

	$= \sin^2\theta/[\cos\theta(\sin\theta-\cos\theta)] - \cos^2\theta/[\sin\theta(\sin\theta-\cos\theta)]$ $= 1/(\sin\theta-\cos\theta) [(\sin^2\theta/\cos\theta) - (\cos^2\theta/\sin\theta)]$ $= 1/(\sin\theta-\cos\theta) \times [(\sin^3\theta - \cos^3\theta)/\sin\theta\cos\theta]$ $= [(\sin\theta-\cos\theta)(\sin^2\theta+\cos^2\theta+\sin\theta\cos\theta)]/[(\sin\theta-\cos\theta)\sin\theta\cos\theta]$ $= (1 + \sin\theta\cos\theta)/\sin\theta\cos\theta$ $= 1/\sin\theta\cos\theta + 1$ $= 1 + \sec\theta\operatorname{cosec}\theta = \text{R.H.S.}$ Therefore, L.H.S. = R.H.S. Hence proved	1 1
29.	(a) Let, $1/\sqrt{x} = a$, $1/\sqrt{y} = b$ the equations are $2a + 3b = 2 \dots (1)$ $4a - 9b = -1 \dots (2)$ Multiply equation (1) by 3 to make the coefficients of b in both equations multiples of each other $6a + 9b = 6 \dots (3)$ Add equation (2) and equation (3) to eliminate b $(4a - 9b) + (6a + 9b) = -1 + 6$ $10a = 5$ Solve for a $a = 5 / 10 = 1/2$ Substitute the value of a into equation (1) to solve for b $2(1/2) + 3b = 2$ $1 + 3b = 2$ $3b = 1$ $b = 1/3$ So, $a = 1/\sqrt{x} = 1/2$, $b = 1/\sqrt{y} = 1/3$	1 1 1

$x = 4, y = 9$ OR (b) Let the fraction be x/y. According to the question, $(x+2)/(y+2) = 9/11$ $11x + 22 = 9y + 18$ $11x - 9y = -4 \dots\dots\dots (1)$ $(x+3)/(y+3) = 5/6$ $6x + 18 = 5y + 15$ $6x - 5y = -3 \dots\dots\dots (2)$ From (1), we get $x = (-4+9y)/11 \dots\dots\dots (3)$ Substituting the value of x in (2), we get $6(-4+9y)/11 - 5y = -3$ $-24 + 54y - 55y = -33$ $-y = -9$ $y = 9 \dots\dots\dots (4)$ Substituting the value of y in (3), we get $x = (-4+9 \times 9)/11 = 7$ Hence the fraction is $7/9$.	1 1 1
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30.



From the figure, we can conclude a few points, which are

(i) $DR = DS$

(ii) $BP = BQ$

(iii) $AP = AS$

(iv) $CR = CQ$

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Since they are tangents on the circle from points D, B, A, and C, respectively.

Now, adding the LHS and RHS of the above equations, we get,

$$DR + BP + AP + CR = DS + BQ + AS + CQ$$

By rearranging them, we get,

1

$$(DR + CR) + (BP + AP) = (CQ + BQ) + (DS + AS)$$

By simplifying,

$$AD + BC = CD + AB$$

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31.

(a)

Area of land (in hectares)	No. of families	
1 – 3	20	
3 – 5	45	f_0
Modal class → 5 – 7	80	f_1
7 – 9	55	f_2
9 – 11	40	
11 – 13	12	

∴ Modal class = 5 – 7, $l = 5$, $h = 2$

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) h = 5 + \left(\frac{80 - 45}{2(80) - 45 - 55} \right) 2 = 6.166\ldots$$

Hence, modal agriculture holdings of the village is 6.17 hectare (approx.)

OR

(b)

Class interval	f_i	x_i (Mid-value)	$d_i = \frac{x_i - 30}{h}$	$f_i d_i$
0-20	7	10	-1	-7
20-40	p	30	0	0
40-60	10	50	1	10
60-80	9	70	2	18
80-100	13	90	3	39
Total	39 + p			60

Assumed mean(A) = 30, Width of the interval (h) = 20

$$\text{Mean} = 30 + \frac{60}{39+p} \times 20 = 54 \Rightarrow 50 = 39 + p \Rightarrow p = 11$$

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Section D

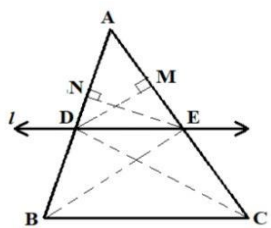
32.	Distance = 360 km Consider x as the speed, then the time taken $t = 360/x$ If the speed is increased by 5 km/h, the speed will be (x + 5) km/h. Distance will be the same. $t = 360/(x + 5)$ We know that Time with original speed – Time with increased speed = 1 $360/x - 360/(x + 5) = 1$ $\text{LCM} = x(x + 5)$ $[360(x + 5) - 360x]/x(x + 5) = 1$ $360x + 1800 - 360x = x(x + 5)$ $x^2 + 5x = 1800$ $x^2 + 5x - 1800 = 0$ $x^2 + 45x - 40x - 1800 = 0$ $x(x + 45) - 40(x + 45) = 0$ $(x - 40)(x + 45) = 0$ $x = 40 \text{ km/hr}$ As we know, the value of speed cannot be negative. Therefore, the speed of the train is 40 km/h.	1 1 2 1
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33.

Given: In $\triangle ABC$, a line l drawn parallel to side BC intersects AB and AC at D and E respectively.

To prove: $\frac{AD}{DB} = \frac{AE}{EC}$

Construction: Draw perpendicular from D and E to AC and AB i.e., $DM \perp AC$ and $EN \perp AB$. Join DC and BE .



Proof:

$$\frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{\frac{1}{2}(AD)(EN)}{\frac{1}{2}(BD)(EN)} = \frac{AD}{DB} \dots\dots\dots(1)$$

$$\frac{ar(\triangle ADE)}{ar(\triangle CED)} = \frac{\frac{1}{2}(AE)(DM)}{\frac{1}{2}(EC)(DM)} = \frac{AE}{EC} \dots\dots\dots(2)$$

Also, $ar(\triangle BDE) = ar(\triangle CED) \dots\dots\dots(3)$

(Triangles on same base and between same parallel are equal in area)

From (1), (2) & (3), we get $\frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{ar(\triangle ADE)}{ar(\triangle CED)}$

$\Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$ (Hence proved)

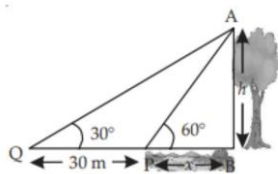
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34.



(a)

Let $AB (= h \text{ m; say})$ be the tree and $PB (= x \text{ m})$

be the river.

By the question,

$\angle APB = 60^\circ$ and $\angle AQB = 30^\circ$ and $PQ = 30 \text{ m}$

In $\triangle PAB$, $\tan 60^\circ = \frac{AB}{PB}$

$$\Rightarrow \sqrt{3} = \frac{h}{x}$$

$$\Rightarrow h = \sqrt{3}x \dots (1)$$

In $\triangle QAB$, $\tan 30^\circ = \frac{AB}{QB}$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{30 + x}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{\sqrt{3}x}{30 + x}$$

$$\Rightarrow 30 + x = 3x$$

$$\Rightarrow 2x = 30$$

$$\Rightarrow x = 15.$$

Putting in (1), $h = \sqrt{3}(15) = 15\sqrt{3} =$

$$15(1.732) = 25.98.$$

Hence, the height of the tree = 25.98 m and

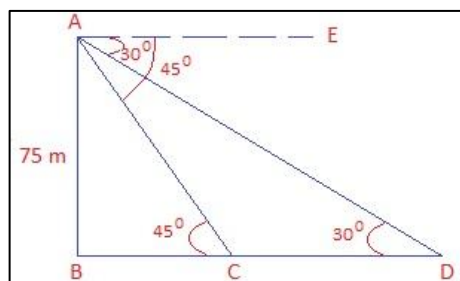
width of the river = 15 m.

OR

(b) Let AB be the lighthouse of height 75 m. Let C and D be the positions of the ships.

30° and 45° are the angles of depression from the lighthouse.

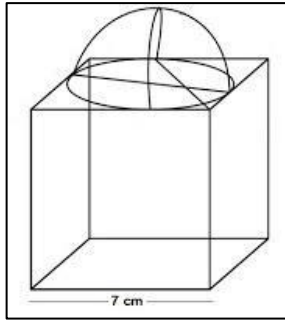
Draw a figure based on given instructions:



	To Find: CD = distance between two ships Step 1: From right triangle ABC, $\tan 45^\circ = AB/BC$ $1 = 75/BC$ $BC = 75 \text{ m}$ Step 2: Form right triangle ABD, $\tan 30^\circ = AB/BD$ $1/\sqrt{3} = 75/BD$ $BD = 75\sqrt{3}$ Step 3: To find measure of CD, use results obtained in step 1 and step 2. $CD = BD - BC = (75\sqrt{3} - 75) = 75(\sqrt{3}-1)$ The distance between the two ships is $75(\sqrt{3}-1)$ m	1 1 1 2 1
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35.

(a) It is given that each side of the cube is 7 cm. So, the radius will be $7/2$ cm.



We know,

The total surface area of solid (TSA) = surface area of the cubical block + CSA of the hemisphere – Area of the base of the hemisphere

$$\therefore \text{TSA of solid} = 6 \times (\text{side})^2 + 2\pi r^2 - \pi r^2$$

$$= 6 \times (\text{side})^2 + \pi r^2$$

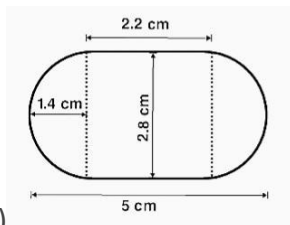
$$= 6 \times (7)^2 + (22/7) \times (7/2) \times (7/2)$$

$$= (6 \times 49) + (77/2)$$

$$= 294 + 38.5 = 332.5 \text{ cm}^2$$

So, the surface area of the solid is 332.5 cm^2

OR



(b)

It is known that the gulab jamuns are similar to a cylinder with two hemispherical ends.

So, the total height of a gulab jamun = 5 cm.

Diameter = 2.8 cm

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	So, radius = 1.4 cm $\therefore$ The height of the cylindrical part = 5 cm – (1.4 + 1.4) cm = 2.2 cm Now, the total volume of one gulab jamun = Volume of cylinder + Volume of two hemispheres $= \pi r^2 h + (4/3) \pi r^3$ $= 4.312\pi + (10.976/3) \pi$ $= 25.05 \text{ cm}^3$ We know that the volume of sugar syrup = 30% of the total volume So, the volume of sugar syrup in 45 gulab jamuns = $45 \times 30\% (25.05 \text{ cm}^3)$ $= 45 \times 7.515 = 338.184 \text{ cm}^3$	1 1 2
	<u>Section E</u>	

36.	(i) $x = 100 - (30 - 32 - 24 - 4) = 10$ (ii) $P(\text{selected person to have Rhesus negative blood type}) = \frac{10+8+6+1}{100}$ $= \frac{25}{100}$ or $\frac{1}{4}$ (iii) (A) $P(\text{person is Rhesus positive but neither A nor B type blood}) = \frac{30+3}{100}$ $= \frac{33}{100}$ OR (iii) (B) $P(\text{person is neither universal recipient nor universal donor})$ $= 1 - \frac{(3+10)}{100}$ $= 1 - \frac{13}{100}$ $= \frac{87}{100}$	1 1 1 + 1 1 + 1
37.	(i) Area of sector = $\frac{(\text{Arc length} \times \text{radius})}{2}$ (ii) Area of sector = $\frac{80}{360} \pi \times 441 = 98\pi \text{ m}^2$ (iii) (A) $\frac{80}{360} \pi \times 441 = \frac{\theta}{360} \pi \times 28^2$ $\theta = 45^\circ$ OR (iii) (B) Increase in the area of the lawn watered = $\frac{80}{360} \pi \times (784 - 441)$ $= 239.56 \text{ m}^2$	1 1 1 + 1 1 + 1
38.	(i) C is nearest to the office. (ii) Distance between A and C $= \sqrt{(5-2)^2 + (3-8)^2} = \sqrt{3^2 + 5^2} = \sqrt{9+25} = \sqrt{34}$ units (iii) (A) $AB = \sqrt{(7-2)^2 + (7-8)^2} = \sqrt{26}$ units $OA = \sqrt{2^2 + 8^2} = 2\sqrt{17}$ units $BC = \sqrt{(7-5)^2 + (7-3)^2} = 2\sqrt{5}$ units $\therefore BC$ is the least. OR (B) Mid point of B & C = D Coordinates of $D = \left(\frac{5+7}{2}, \frac{3+7}{2} \right) = (6, 5)$	1 1 1 + 1 1 + 1